INCH-POUND
MIL-T-17370C
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(See 6.11)

MILITARY SPECIFICATION

TUBE AND FLUE CLEANER, BOILER, INTERNAL, AIR MOTOR DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers mechanical, air-motor-driven tube and flue cleaners for cleaning the interior of boiler tubes and flues.

1.2 Classification. The tube and flue cleaners covered by this specification shall be of the following types (see 6.2):

Type I - Fire tube.
Type II - Water tube.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

ZZ-H-500 - Hose, Rubber, and Hose Assemblies, Rubber: Pneumatic (Yarn or Fabric Reinforced).
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Semites.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3699 - Standard Specification for Kerosine. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article (type I only). When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.2 Qualification (type II only). The type II air-motor-driven tube and flue cleaners furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.5 and 6.6).

3.3 Component parts. Each tube and flue cleaner for fire- or water-tube boilers, regardless of size, shall be constructed so that it can be joined together in proper order of assembly into one complete unit, which shall consist of the following parts:

(a) Type I cleaner:

(1) Air motor.
(2) Operating hose complete with couplings attached.
(3) Vibrator head assembly.
(4) Universal joint.
(5) Knocker heads (two).
(6) Set of necessary wrenches.
(7) Repair parts list and operating instructions.

(b) Type II cleaner:

(1) Air motor with short hose and coupling attached.
(2) Universal joint (when required)
(3) Flexible holder.
(4) Operating hose complete with couplings attached.
(5) Set of necessary wrenches.
(6) Repair parts list and operating instructions.

3.4 Operating pressure. The tube and flue cleaners shall effectively clean the walls of boiler tubes when operated with compressed air supplied at gauge pressure between 80 and 100 pounds per square inch (lb/in^2).

3.5 Materials. Materials shall be as specified herein. Contact between dissimilar metals susceptible to galvanic action, especially brass, copper, or steel in contact with aluminum or magnesium, shall be avoided. Material shall be free of defects or imperfections that may affect the serviceability of the finished tube and flue cleaner.

3.5.1 Fungal resistance. Use of materials that are nutrients for fungi shall be avoided, Where such materials must be used, they shall be treated with a fungicidal agent acceptable to the contracting activity.

3.5.2 Recovered materials. Unless otherwise specified herein, all equipment and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended

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use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.6 General construction. The tube and flue cleaners specified herein shall be of two types (see 1.2) and shall perform the functions of removing rust, scale, and other normal boiler deposits from the walls of tubes in fire- and water-tube boilers. Parts subject to wear, breakage, or distortion shall be accessible for easy adjustment and repair with no special tools required. Special tools are those tools not listed in the Federal Supply Catalog.

3.6.1 Type I cleaner sizes. The type I tube and flue cleaner shall be sized to clean the walls of straight fire tube boiler tubes as specified in table I.

TABLE I. Size - fire tube units.

Cleaning unit size designation	Outside diameter of boiler tube (inches)	Minimum inside diameter of boiler tube (inches)
1-1/2	1-1/2	1-1/4
2	2	1-3/4
2-1/2	2- 1/2	2- 1/4
3	3	2-5/8
3- 1/2	3- 1/2	3- 1/8
4	4	3-5/8

3.6.2 Type II cleaner sizes. The type II tube and flue cleaner shall be sized to accommodate the brushes used to clean the interior walls of boiler water tubes as specified in table II.

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TABLE II. Size - water tube units (type II).

Cleaning unit size designation	Outside diameter of boiler tube (inches)	Wall thickness of boiler tube (inches)	Dimensions of curved sections		
			Radius (inches)	Arc (degrees)	Maximum crush of tube (inch)
1	1	0.095	5	75	0.050
1 SH 1/	1	.109	2-1/4	180	.0625
1-1/8	1-1/8	.109	5	75	.050
1-1/4	1-1/4	.109	5	75	.062
1-1/4 special	1-1/4	.095 to .134	4	90	.0625
1-1/4 SH 1/	1-1/4	.165	2-1/2	90	.0625
1-1/4 SH 1/	1-1/4	.120	2-1/2	180	.0625
1-3/8	1-3/8	.120	6	75	.062
1-1/2	1-1/2	.120	6	75	.075
1-1/2 SH 1/	1-1/2	.134	4	180	.0625
1-1/2 special 1/	1-1/2	.105 to .120	2-1/2 to 5	75 to 180	.0625
1-3/4	1-3/4	.134	6	75	.100
2	2	.134	6	75	.120
2 SH 1/	2	.148	5	90	.120
2 SH 1/	2	.165	2-3/4	180	.156
2-1/2	2-1/2	.165	12	90	.120
3	3	.165	12	75	.125
3A 2/	3	.250	12	75	.125
	4	.250	15	75	.125
3-1/4 2/	3-1/4	.250	18	75	.125
	4-1/2	.343	16	75	.125
3-1/4A	3-1/4	.375 max	10	62	.125
4	4	.375 max	15	120	.125
4	4	.375 max	10	53	.125
4-1/2	4-1/2	.343 max	15	48	.125
4-1/2	4-1/2	.343 max	12	90	.125

1/ Special units shall be constructed for special superheater (SH), side or rear wall, roof or economizer tube configurations, and header conditions.
2/ Constructed for both tube sizes specified.

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3.6.3 Oversize and special cleaners, both tries. Tube cleaners of sizes larger than 4-1/2 inches shall be as approved by the contracting activity (see 6.2). Sizes designed for special boiler tube conditions shall be qualified and approved in accordance with 4.5 for the specific application intended. These special units may be evaluated by operation in a test boiler, or in the specific boiler for which intended, and will not be accepted by the Government until proper testing has been approved.

3.7 Castings and forgings. Castings shall be free of visible blowholes, porosity, hard spots, shrinkage defects, cracks, and other defects. Forgings shall be free from scale, enclosures, cold shuts, mismatching, and other defects that may affect their structural strength. Castings and forgings shall have sufficient strength and other physical and chemical properties to meet the performance requirements specified herein.

3.8 Welding, brazing, and soldering. Welding, brazing, and soldering shall be performed in accordance with commercial practice.

3.9 Fastening devices. Screws, pins, bolts, and similar parts shall be provided with means for preventing loss of tightness and adjustment. Parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed. Unless otherwise approved by the contracting activity, steel pins used in universal joints shall be riveted. Devices to be screwed to motors shall have fastenings or threading such that rotation of the motor tightens the connection.

3.10 Standard hardware parts. Standard ASTM or ASME hardware parts (such as screws, nuts, bolts, and studs) shall be used.

3.11 Threads. Dimensions of all threads, tolerances, and gauges for determining interchangeability shall comply with FED-STD-H28. Tapered pipe threads shall be used where applicable. Thread class selection shall be unified coarse (UNC) class 2A-2B for general use and UNC class 3A-3B for tighter fits.

3.12 Lubrication. Bearings (except sealed-for-life type) and other working parts shall be provided with means for lubrication.

3.13 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.14 Performance.

3.14.1 Hose performance. Hose shall be tested to hydrostatic pressure of 250 lb/in^2 and shall show no leakage under test (see 4.7.3).

3.14.2 Motor performance. Motors shall have performance characteristics superior or equal to those shown in table III for type I cleaners or table IV for type II cleaners (see 4.7.4).

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TABLE III. Performance characteristics of motors for fire-tube units.

Cleaning unit size designation (inches)	Performance at specified torque		
	Specified torque (foot-pounds) (minimum)	Speed (r/min) (minimum)	Air consumption (ft^3/min per horsepower) (maximum)
1-1/2	0.09	4000	200
2	.40	4000	175
2-1/2	1.50	3000	125
3	2.75	3000	100
3-1/2	3.50	3000	75
4	3.50	3000	75

TABLE IV. Performance characteristics of motor for water-tube units.

Cleaning unit size designation (inches)	Performance at specified torque		
	Specified torque (foot-pounds) (minimum)	Speed (r/min) (minimum)	Air consumption (ft^3/min per horsepower) (maximum)
1	0.03	4000	300
1 SH	.03	4000	300
1-1/8	.04	4000	275
1-1/4	.06	4000	250
1-1/4 SH	.06	4000	250
1-3/8	.07	4000	225
1-1/2	.09	4000	200
1-1/2 SH	.09	4000	200
1-3/4	.12	4000	175
2	.14	4000	150
2 SH	.14	4000	150
2-1/2	.19	4000	125
3	1.20	4000	100
3A	.65	4000	100
3-1/4	.65	4000	100
3-1/4	1.20	4000	100
4	2.00	4000	60
4-1/2	2.50	4000	50
5	3.00	4000	40
6	4.00	3500	40
8	6.00	3000	40

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3.15 Components.

3.15.1 Hose.

3.15.1.1 Construction. Hose for all units shall be in accordance with ZZ-H-500.

3.15.1.2 Reinforcement. Hose for the size 1-1/2 units and larger shall be reinforced externally with 5/32- by 5/64-inch half-round wire, wound left hand, with turns spaced on centers of not more than 3/4 inch. The wire winding shall terminate and be secured in or immediately behind the hose couplings, without protrusion of wire ends beyond the outside surface of the couplings.

3.15.1.3 Couplings. Hose couplings shall be as required (see 6.2). Couplings shall conform to tables V and VI.

TABLE V. Hose and coupling sizes - type I cleaning unit.

Cleaning unit size designation (inches)	Nominal inside diameter of hose (inch)	Coupling pipe thread size male (inch)
1-1/2	3/8	3/8
2	1/2	1/2
2-1/2	3/4	3/4
3	1	1
3-1/2	1	1
4	1	1

TABLE VI. Hose and coupling sizes - type II cleaning units.

Cleaning unit size designation (inches)	Nominal inside diameter of hose (inch)	Length over couplings (short hose only) (inches)	Coupling pipe thread size (inch)
1 and 1 SH	1/4	5 to 5-1/2	1/4
1-1/8	1/5	5-1/4 to 5-3/4	1/4
1-1/4 and 1-1/4 SH	1/4	5-3/8 to 5-7/8	1/4
1-3/8	3/8	5-3/8 to 5-7/8	3/8
1-1/2 and 1-1/2 SH	3/8	5-1/2 to 6	3/8
1-3/4	3/8	5-5/8 to 6-1/8	3/8
2 and 2 SH	3/8	5-3/4 to 6-1/4	3/8
2	1/2	5-3/4 to 6-1/4	1/2
2-1/2	1/2	13-1/4 to 14-1/4	1/2
3	1	9-1/4 to 10	1
3	3/4	13-1/4 to 14-1/4	3/4
3A	1/2	13-1/4 to 14-1/4	1/2
3-1/4	1/2	13-1/4 to 14-1/4	1/2
4	3/4	13-1/4 to 14-1/4	3/4
4-1/2	1	13-1/4 to 14-1/4	1

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3.15.1.4 Short hose (type II only). The short hose for connecting to the motor shall be fitted with a female pipe thread coupling on the air supply end for attachment to the (long) operating hose.

3.15.1.5 Long hose (types I and II). The long hose for connecting the cleaning unit to the compressed air line shall be fitted at each end with a male threaded coupling. Unless otherwise specified (see 6.2), hose for type I cleaners shall be 50 feet long, hose for type II cleaners shall be 25 feet long. Tolerance for each length shall be plus or minus 2 inches.

3.15.2 Motor.

3.15.2.1 Motor construction. Motors shall be simple in construction, with parts readily accessible for adjustment or replacement. The working parts exposed to wear shall be of sufficient hardness and toughness to withstand the service required, and shall be finished to dimensions within standard limits. All parts of motors shall be interchangeable and of good fit when assembled with replacement parts for the same size, type, and make. Motors shall be furnished in condition to develop maximum efficiency without necessity of preliminary operation to improve working fit of parts. The motor shaft shall have a male thread and shall be provided with flat surfaces for engagement with an open-end wrench. The operating compressed air shall be exhausted in such manner that refuse from the tubes is blown ahead of the cleaning device.

3.15.2.2 Motor fitting (type I cleaner). Motors shall be equipped at the inlet end with a female pipe thread.

3.15.2.3 Motor fitting (type II cleaner). Motors shall be equipped at the inlet end with a section of short hose conforming to 3.15.1.4.

3.15.3 Flexible holder (type II cleaner).

3.15.3.1 Construction. Flexible holders shall be of durable construction with the ends of the flexible element secured to the terminal fittings.

3.15.3.2 Terminal fittings. Terminal fittings on flexible holders shall be provided with flat surfaces for engagement with open-end wrenches. The fitting on the end toward the motor shall be threaded internally to allow direct attachment to the motor shaft, except where a universal joint is used between the motor shaft and flexible holder. Where a universal joint is used, the fitting on the end of the flexible holder toward the universal joint shall be threaded externally to allow direct attachment to it. The fitting on the end toward the cleaning device shall be threaded and shall conform to table VII.

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TABLE VII. Flexible holder for type II cleaner.

Cleaning unit size designation	Threads cleaning device end			Length (inch)	Overall length (minimum) (inches)
	Size (outside diameter) (inch)	Threads per inch 1/	Type		
1	5/16	18	Female	5/16 to 3/8	3- 1/2
1 SH	5/16	18	Female	5/16 to 3/8	8
1-1/8	5/16	18	Female	5/16 to 3/8	3-5/8
1- 1/4	5/16	18	Female	5/16 to 3/8	3-5/8
1-1/4 SH	5/16	18	Female	5/16 to 3/8	8
1-3/8	5/16	18	Female	5/16 to 3/8	3-5/8
1-1/2	5/16	18	Female	5/16 to 3/8	4- 1/4
1-1/2 SH	5/16	18	Female	5/16 to 3/8	8
1-3/4	7/16	14	Female	3/8 to 7/16	6
2	7/16	14	Female	3/8 to 7/16	6
2 SH	7/16	14	Female	3/8 to 7/16	9
3	9/16	12	Male	7/16 to 1/2	9-1/8
3A	9/16	12	Male	7/16 to 1/2	9- 1/8
3- 1/4	9/16	12	Male	1/2 to 9/16	9-1/8
4	9/16	12	Male	1/2 to 9/16	10-9/16
4-1/2	7/8	9	Male	9/16 to 5/8	10-9/16

1/ Tolerance UNC 2A-2B applies; UNC 3A-3B acceptable.

3.15.3.3 Length. Overall length of flexible holders with fittings attached shall conform to table VII. Flexible holders 8 inches or more in length shall have a protector ring fastened onto the flexible unit midway between the terminals.

3.16 Carrying case. The tube and flue cleaner, with repair parts, but without the long hose, shall be furnished in a wooden or metal tool kit or box, The kit or box shall be provided with a pair of substantial hinges and a hasp or lock on the cover, and shall be equipped with a metal carrying handle in the center of the cover. The exterior of the carrying case shall be painted or varnished to produce a neat and durable surface. The interior of metal carrying cases shall be painted or otherwise treated to resist rust or corrosion.

3.17 Identification marking. Identification marking shall be in accordance with MIL-STD-130, as practicable. As a minimum, each part shall be permanently and legibly marked with the manufacturer's part number. The motor in each unit shall be permanently marked to show plainly the manufacturer's model, type, and size designation, together with the manufacturer's name or trademark of such known character that the source of manufacture can be readily determined. Where permanent marking is not practicable, legible marking by tag or affixed tape is acceptable.

3.18 Workmanship. Standards shall be such that equipment specified herein shall conform to the requirements in section 3. All components, subassemblies, parts, and final assembly shall withstand the tests in section 4.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Qualification inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions of MIL-STD-105.

4.4 First article inspection (type I only). First article inspection for type I cleaners shall consist of the tests specified in 4.7.1, 4.7.3, 4.7.4 through 4.7.4.4 and the visual and dimensional examinations as specified in 4.6.2 and 4.6.3. First article sampling shall consist of one type I cleaner. Failure of the first article examination shall be cause for rejection (see 6.3).

4.5 Qualification inspection (type II only). Qualification inspection for type 11 cleaners shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification inspection shall consist of the tests specified in 4.7.2 through 4.7.4.4.

4.6 Quality conformance inspection,

4.6.1 Inspection lot. All cleaning units of the same size, class, and type presented at one time shall be Considered a lot for purposes of quality conformance inspection and tests.

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4.6.2 Sampling for examination. A random sample of units shall be selected from each lot of material in accordance with table VIII for examination of visual and dimensional characteristics.

TABLE VIII. Sampling for examination.

Number of units in inspection lot	Number of units in sample	Acceptance number (defective)	Rejection number (defective)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1

4.6.3 Examination. Each of the sample cleaning units selected in accordance with 4.6.2 shall be subjected to surface examination for all the requirements of this specification not involving tests. Examination shall be conducted as specified in table IX. Any cleaning units in the sample containing one or more defects shall not be offered for delivery.

TABLE IX. Classification for defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Evidence of use of unauthorized material.
102	Unit not complete, component parts missing; not nominal size specified.
103	Hose not ply construction specified; inside diameter length over coupling nonconforming.
104	Hose not reinforced externally; reinforcing wire not wound left hand; turns not spaced on centers as specified; wire winding not terminated immediately behind hose coupling; not free protruding.
105	Short hose not fitted with female pipe thread coupling on air supply end (type II).
106	Long hose not fitted at each end with male threaded couplings; length not within tolerance specified.
107	Hose couplings not provided with wrench flats; threads not type, size, or class of fit specified; not in accordance with FED-STD-H28.
108	Flexible element not secured in terminal fittings (type II).
109	Flexible holder less than minimum length specified (type II).
110	Terminal fittings not provided with wrench flats (type III).
111	Threads on flexible holder terminal fittings not fit specified; not in accordance with FED-STD-H28 (type II).

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TABLE IX. Classification for defects - Continued.

Categories	Defects
Major:	
112	Threads (cleaning device end) not of diameter, number per inch, type, or length specified (type II).
113	Fitting on universal joint end not threaded internally to match motor shaft threads (if applicable).
114	Fitting on universal joint end not threaded externally; threads do not mate with universal joint (if applicable).
115	Motor parts not accessible for adjustment or replacement.
116	Parts not interchangeable with replacement parts for same size, type, and make.
117	Motor shaft not provided with male thread; not provided with wrench flats.
118	Inlet end of motor not equipped with section of hose (type II motor only).
119	Burrs and rough edges of metal parts not removed.
Minor:	
201	Motor identification marking missing; data incomplete; not legible; not permanent.
202	Carrying case not provided with hinges, hasp, or hook, or metal handle in cover.
203	Exterior and interior of case not finished in accordance with specification.

4.6.4 Sampling for tests. One-fifth of the units selected for the examination specified in 4.6.3 shall be tested in accordance with 4.7.3 and 4.7.4.

4.6.5 Possible test failures. Possible test failures are defined as follows:

(a) Performance test:

(1) Cleaners (except SH) will not pass through specified fire- or water-tube sections.
(2) Evidence that units do not clean effectively in specified straight and curved tube sections.
(3) SH units will not pass through and clean in specified straight tube sections (type II).
(4) Motor shaft rotation is not clockwise.

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(b) Motor tests (under specified conditions):

(1) Air consumption exceeds upper limit.
(2) Speed requirements not met.
(3) Torque requirements not met.
(4) Final speed less than that required.
(5) Refuse from tube cleaning is not blown ahead of the cleaning device.

(c) Hose hydrostatic tests (specified time and pressure):

(1) Evidence of hose leakage.

4.7 Test procedures.

4.7.1 Type I cleaner performance. Each size of completely assembled tube and flue cleaner shall be tested for satisfactory operation in its corresponding size of straight fire tube as specified in table I. The straight tubes used on the test stand shall be actual tubes for fire-tube boilers. Not more than 15 minutes of the operating time shall be allocated to checking operation and fit in fire tubes. Special sizes of cleaners not included in table I shall be constructed for the use intended and shall be at least equal in performance to the next smaller size.

4.7.2 Type II cleaner performance.

4,7.2.1 First test. Each size of completely assembled type II cleaner, including a standard cleaning device (expanding brush) from stock, shall be tested for satisfactory operation in corresponding sizes of bent tubes conforming to dimensions specified in table II. Unless otherwise specified (see 6.2), each size of cleaner, except sizes of superheater and economizer tubes, when completely assembled to include motor hose, flexible holder, and a cleaning device, shall pass through and effectively operate in straight and tuned sections of sizes of boiler tubes specified in table II. Cleaners for use in superheater and economizer tubes shall operate in the straight section of the tubes, but will not be required to negotiate the tuned section specified in table II. Special sizes of units not included in table II shall be constructed for the purpose intended, and shall be at least equal in performance to the next smaller size. This portion of the test shall be conducted on boiler tubes that are at least 12 feet in length, and have at least two bends (one at each end) as specified in table II for the size being tested.

4.7.2.2 Second test. The assembled cleaner shall be supplied with air at 80 to 100 lb/in^2 gauge pressure and shall be used to clean dirty boiler tubes for a period of 5 hours without malfunction or excessive loss of horsepower. The dirty tubes shall be tubes actually installed in water-tube boilers scheduled for cleaning. The cleaning equipment shall be handled by the usual personnel assigned to boiler maintenance. For purposes of comparison, qualified tube and flue cleaners of the same size, drawn from stock, shall be operated by the boiler maintenance personnel on the same cleaning job during the same time period.

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4.7.3 Hydrostatic test. Hose for all sizes of units, with couplings attached, shall withstand a hydrostatic pressure of 250 lb/in^2 maintained for 2 minutes, without exposing imperfections or leaks in either the hose or the couplings.

4.7.4 Motors. Motors shall be tested for horsepower and air consumption when operating with continuous lubrication introduced with the air supply at the rate of 10 drops per minute for size 1 to 1-1/2 cleaners and 20 drops per minute for all larger sizes. The lubricant shall consist of three parts kerosene and one part lubrication oil by volume. The kerosene shall conform to ASTM D 3699. The lubricating oil shall conform to Military symbol 2110-T-H of MIL-H-17672.

4.7.4.1 Performance of motors. Motors subjected to tests shall show performance characteristics at least equal to those shown in table III for type I motors, or table IV for type II motors. The final speed of any motor at the end of the 10-minute test specified in 4.7.4.4 shall be not less than one-half the speed shown in these tables for the corresponding type and size of motor.

4.7.4.2 Horsepower test. Compliance of motors shall be determined by horsepower tests made by means of a mechanical brake or an electrical dynamometer of suitable capacity. Horsepower shall be based on operation of the motor with compressed air supplied at a gauge pressure of 90 lb/in^2, measured at the point where the longer length of hose furnished with the outfit is attached to the compressed air line. Horsepower shall be computed by the following formula:

$$\text{Horsepower} = 2\pi RLN/33000$$

Where :

R = Length of radius or lever arm through which torque load is applied, measured in feet.
L = Load applied at radius R, in pounds-force.
N = Speed, in revolutions per minute.

4.7.4.3 Air consumption test. Air consumption shall be determined with a flow meter of an approved type. Air consumption shall be expressed as standard cubic feet per minute, based on operation with compressed air supplied at a gauge pressure of 90 lb/in^2.

4.7.4.4 Speed test. Speed data shall be obtained with a tachometer. Motors shall also be tested with the same equipment and lubricant as specified in 4.7.4, except that for sizes of 2 inches and smaller, 1 milliliter of lubricant shall be introduced into the short length of hose fitted on the motor or into the operating hose at the motor end as applicable, while the motor is at rest. For 2-1/2 inch size cleaners and larger, 5 milliliters of lubricant shall be used. The motor shall then be operated and the specified torque load immediately applied. Additional lubricant shall not be supplied during the test. The speed of the motor shown after operation for 10 minutes under the applied load shall be recorded as the final speed.

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4.8 Inspection of packaging. Sample packages and packs, and the inspection of the presentation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7).

5.1 General.

5.1.1 Navy fire-retardant requirements. Navy fire-retardant requirements shall be as specified below:

(a) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(b) Fiberboard. Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, cleated fiberboard boxes including interior packaging forms shall meet the flame spread index and the specific optic density requirements as specified in PPP-F-320.

5.2 Preservation, packing, and marking. Tube and flue cleaner shall be preserved level A or commercial and packed level A, B, C, or commercial as specified (see 6.2) in accordance with PPP-P-40. Packing of level A, B, and C shall be in accordance with table IV of PPP-P-40, except that level B shipping containers shall be of the weather resistant type. Each tube and flue cleaner shall be packaged as a complete unit (see 3.3). Unit packs, intermediate packages, and shipping containers, in addition to any special markings required by the contract or purchase order (see 6.2), shall be marked in accordance with PPP-P-40.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Air-motor-driven tube and flue cleaners are intended for use in cleaning internal surfaces of boiler tubes, condenser tubes, and other heat exchanger tubes.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.1).
(e) Tube cleaner sizes if larger than 4-1/2 inches (see 3.6.3).
(f) Type of coupling required (see 3.15.1.3).
(g) Length of hose required, if other than specified (see 3.15.1.5).
(h) If inspections are not to be performed in accordance with MIL-STD-105 (see 4.3).
(i) If cleaners need not operate in straight and tuned tubes (see 4.7.2.1).
(j) When fire-retardant materials are not required (see 5.1.1).
(k) Level of preservation and packing required (see 5.2).
(l) Special marking required (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-T-4902	First article inspection report	----

The above DID’s were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID’s are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

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6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List No. 17370 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests must be made in accordance with "Provisions Governing Qualification SD-6" (see 6.6.1).

6.6.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.8.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

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6.9 Special provisions for curved sections of water tubes (type II cleaners only). The requirements specified in 3.6.2 are for operation in curved sections of water tubes and take account of deformation of the tube, which results in reduction of the inside diameter of the tube at one or more points in the curve, and which is designated as crush. Where the dimensions of curved sections of tubes for which type II cleaners are required differ from the dimensions specified in table II, columns 4, 5, and 6, detailed dimensions of the bend should be specified in the contract or purchase order.

6.10 Subject term (key word) listing.

Air consumption
Couplings
Fire tube
Torque
Universal joint
Water tube

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 4940-0560)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-17370C	TUBE AND FLUE CLEANER, BOILER, INTERNAL, AIR MOTOR DRIVEN

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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